

Uday Singh Kanwar

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[Person Website](#)

[Github](#)

Education

McMaster University

Expected Graduation, April 2027

Bachelor of Engineering (Software Engineering)

Hamilton, Ontario

- GPA: 3.9/4.0- Dean's Honor List, **Faculty of Eng Award of Excellence Scholarship**- \$4000

Related Coursework

Sep 2024 - Present

- Linear Algebra, OOP (Java), SWE Basics (Linux, Git Version Control, CI Pipeline, C), Digital Logic (Verilog, FPGA), Discrete Math, Statistics, Design Projects (Python, CAD), Computer Architecture, Data Structures and Algorithms

Google Machine Learning Course

Nov 2024

- Linear Regression, Logistic Regression, Neural Networks, LLM's, Handling Datasets, Over-Fitting, Embeddings

Experience

Full-Stack Developer

Sep 2024 - Dec 2024

Human Endeavour

Remote, part-time

- Managed **MySQL database** that handles real-time sensor data, optimizing query performance by designing **indexing structures** and implementing efficient data retrieval methods.
- Created a **Figma** UI/UX model and Developed a **React**-based front-end for visualizing sensor **data**, also utilized **Tableau** for creating dynamic and real-time dashboards for **data analytics**.

Embedded Software Engineer Intern

June 2024 - Aug 2024

Human Endeavour

Vaughan, Ont

- Developed a **C++** fall detection **algorithm** using an ESP32 micro-controller and accelerometer to monitor fall events with **95% accuracy** and implemented an automated email alert service via the **Gmail SMTP server**.
- Implemented a **trilateration algorithm** with Ultra-Wideband technology using **AT Commands** and **C++** for an indoor positioning system with **80% accuracy** in tracking.
- Debugged **100+ ESP8266 PCBs** and configured **Wi-Fi** communication between the chips and a MySQL database.

Projects and Extra-Curriculars

ML-Powered Fraud Detection | Python, Scikit-learn, Tensorflow, Pandas, NumPy, Matplotlib, Seaborn, Kaggle

Dec 2024

- Evaluated different models such as **Logistic Regression**, **SVM**, **KNears Neighbours**, **Decision Trees** for classifying fraudulent credit card transactions in an imbalanced dataset.
- Achieved **98.7%** (SMOTE Over-Sampling), and **94.21%** (Random Under-Sampling) with Logistic Regression.
- Implemented a **Neural Network** with **Tensorflow** and achieved validation accuracy of **92.1%**
- Performed feature scaling, outlier removal, correlation analysis, data **visualization** and **analytics**.

Bus Forecast @BramHacks | Typescript, Python, Flask, Google Maps API, Facebook Events API, React.js, Vite

Nov 2024

- **Won**: Developed a **predictive** model for public transportation surges in city of Brampton, using Google Maps API, Facebook Events API, and Census data to **optimize** bus deployment and enhance service efficiency.

EcoCoin @Terrahacks | JavaScript, React, Firebase, CSS, HTML

Aug 2024

- Developed EcoCoin, an eco-friendly **cryptocurrency**, and EcoHub, a platform to track and reward environmental contributions using React, Firebase, Mapbox, and Google Cloud.

Memories @Makeuoft | Python, OpenCV, C++, Arduino IDE, Taipy, Tensorflow (ML)

Feb 2024

- Built a **computer vision** program in Python for people with dementia using **OpenCV** for live facial recognition, integrated into a website with **Taipy**, and connected to an **Arduino** board using **pyserial** and C++.

Teacher Announcement Program | JavaScript, HTML, CSS

Sep 2021 - Dec 2021

- Developed a fully **responsive website** for teachers to post announcements that students can browse. It uses a browser's **local storage** to store the posted announcements and includes a comprehensive **search algorithm**

Hobbies and Clubs | Google Student Developer's Club, Intramurals: Hockey, Basketball and Volleyball

Sep 2023 - Present

Skills

Languages: Python, Java, JavaScript, C++, C, HTML, CSS, Arduino, C, MS Power Apps, React.js, TypeScript

Technologies: Pandas, NumPy, Matplotlib, Seaborn, Kaggle, Scikit-learn, Firebase, Flask, phpMyAdmin, TensorFlow, SQL, OpenCV, Taipy, Arduino IDE, MatLab, Git, GitHub, AutoDesk Inventor